# Roy Ho

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## EDUCATION

**UNIVERSITY OF CALIFORNIA, DAVIS**
**Bachelor of Science** **September 2022 – March 2026**
Major in Statistical Data Science and Minor in Computer Science

**Relevant Coursework:** Statistical Learning, Regression, Probability Theory, Advanced Computing, Algorithm Design & Analysis, Data Structures

**Certifications:** NVIDIA – Fundamentals of Deep Learning **October 2025**

## EXPERIENCE

**XLR8 AI**
**AI Search Strategist** **November 2026 – Present**
- Design query sets as evaluation suites measuring how LLM-based search engines (ChatGPT, Perplexity, Gemini, Google AI Overviews) retrieve and cite client brands, running controlled experiments and analyzing citation patterns to diagnose visibility gaps.
- Implement technical fixes making client sites machine-readable for AI crawlers: JSON-LD schema, llms.txt, robots.txt, sitemap.xml, metadata.
- Onboard clients and translate experiment results into prioritized technical and content recommendations presented in client reviews.

**Journal of Artificial Intelligence and Knowledge Engineering (JAIKE)**
**Artificial Intelligence Researcher** **January 2025 – Present**
- Conducted research on LLM-based automation covering API-driven system design, agent orchestration frameworks, retrieval-augmented generation (RAG), and a consolidated threat model spanning prompt injection, data leakage, and unintended action execution.
- First-authored a 30-page, 36-source survey paper introducing a two-axis framework for LLM agent autonomy and oversight, plus a cost model pricing oversight against productivity gains; revise-and-resubmit with publication recommended.
- Peer-reviewed JAIKE submissions on LLM retrieval, long-task reasoning, and model architectures for methodological rigor.

**TechSprint Innovators**
**Head of Data Engineering** **March 2024 – September 2025**
- Built a multi-factor stock screening model using fundamental, technical, and NLP-based sentiment features (FinBERT).
- Developed a scikit-learn classification model predicting price appreciation, with feature selection, tuning, and validation on historical market data.
- Engineered and automated a daily ETL (Extract, Transform, Load) data pipeline (Python, yfinance, Alpaca API) running on a Raspberry Pi to filter equities, generate structured CSV outputs, and deliver real-time investment signals via Discord webhook.

## SIGNIFICANT PROJECTS

**San Francisco Restaurant Safety Map** **April 2026 – June 2026**
- Built a full-stack web app mapping 20,000+ health inspections across 7,700+ San Francisco restaurants using the DataSF public feed.
- Engineered a Python Extract, Transform, Load (ETL) pipeline (pandas, geopy) normalizing inspection records into a 3-table SQLite schema with cached geocoding, then containerized the API and frontend with Docker Compose using a multi-stage Node-to-nginx build.
- Developed a Flask REST API with 7 endpoints and a React + Mapbox frontend featuring debounced search, neighborhood typeahead, geolocation-based "Near Me" centering, and an Insights panel surfacing per-neighborhood best/worst rankings via SQL window functions.

**Job Market Analytics Dashboard** **September 2025 – December 2025**
- Owned the data acquisition layer for a group capstone dashboard, writing JobSpy and Selenium scrapers that collected 23,000+ job posting records across seven role families in CA, NY, and TX, deduplicated to 6,800+ unique postings from 2,300+ companies.
- Built the resume-to-job matching engine as a weighted ranking model combining Term Frequency-Inverse Document Frequency (TF-IDF) cosine similarity, skill overlap against a 366-term taxonomy, and an experience-proximity bonus.
- Implemented eligibility pre-filters that stripped senior-level titles and regex-extracted required years of experience from descriptions to drop postings above the candidate's level, then delivered the ranked output into the final dashboard.

**Drowsy Driver Detection System** **January 2025 – March 2025**
- Owned the eye-state detection layer of a group real-time drowsy driver detection project, awarded Best Execution for the AISC Winter 2025 cycle.
- Computed Eye Aspect Ratio from dlib's 68-point facial landmark model and frontal face detector via scipy Euclidean distance, flagging drowsiness when EAR held below a 0.25 threshold across 20 consecutive frames.
- Integrated the detector into an OpenCV pipeline with grayscale preprocessing and convex hull eye overlays, adding a pygame audio alert and graceful failure handling that surfaces setup instructions when model assets are missing.

## SKILLS

**Programming:** Python, R, SQL, JavaScript, HTML, CSS, JSON-LD
**Data Analysis & Machine Learning:** Excel (advanced formulas, pivot tables, VLOOKUP, data visualization), Regression (Lasso, Ridge, Logistic), Random Forest, Gradient Boosting, CNNs, Feature Engineering, Cross-Validation & Hyperparameter Tuning, Model Evaluation (ROC-AUC, Precision/Recall), TF-IDF, NLP (spaCy)
**Libraries & Frameworks:** scikit-learn, pandas, NumPy, PyTorch, Flask, SQLAlchemy, SQLite, React, Vite, Mapbox GL JS, BeautifulSoup, Selenium, JobSpy, PySpark, Matplotlib, geopy, scipy, dlib, OpenCV
**Tools & Platforms:** Tableau, Power BI, Docker, Git, GitHub, Jupyter Notebook, VS Code, Cursor, Raspberry Pi